

Assignment-6

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Gymnasium Environments - cliffWalking - Taxi

$$E = (S, A, R, P)$$

Random action: $A_t \sim \text{uniform}(A)$

If there are n possible actions, then:

$$P(A_t = a) = \frac{1}{n}$$

$$a \in A$$

State Transition:

$$S_t, A_t \rightarrow R_t, S_{t+1}$$

$\therefore (S_{t+1}, R_t) \in \text{Environment}(S_t, A_t)$

Complete interaction is

$$S_t \rightarrow A_t \rightarrow R_t \rightarrow S_{t+1}$$

Total Reward:

$$C_1 = \sum_{t=0}^{T-1} R_t$$

$$R_0 = -1$$

$$R_1 = -1$$

$$C_1 = (-1) + (-1) + (-1) + (-100)$$

$$R_2 = -1$$

$$R_3 = -100$$

$$C_1 = -103$$

$$R_3 =$$

Avg Reward:

$$R_{avg} = \frac{C}{T} = \frac{-103}{4} = -25.75$$

Multiple Episodes:

$$C_{avg} = \frac{1}{N} \sum_{i=1}^N C_i$$

Random Agent Algorithm

$$S_0 = \text{Reset}(t)$$

t : $A_t \sim \text{uniform}(A)$

$$(S_{t+1}, R_t) = \text{step}(S_t, A_t)$$

$$C_t = C_t + R_t$$

Continue until:-

Terminated = True

$$R_{avg} = \frac{C}{T}$$

Cliff Walking & Taxi

$$A = \{0, 1, 2, 3\}$$

$$P(A_t = a) = \frac{1}{4}$$

for Taxi:

$$A = \{0, 1, 2, 3, 4, 5\}$$

$$\text{so: } P(A_t = a) = \frac{1}{6}$$

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